Grade 10 Fundamentals – Mathematical Notation

Contents

Evaluated criteria

C0 Interpret and use basic mathematical notation for quantities, patterns, variables, ellipses, and indexed collections.

1 From Everyday Quantities to Short Names

Problems:

1.1 Lists, Patterns, and Ellipses (C0)

Mina adds \$2 to her jar each day:

Day	1	2	3	4	5
Total saved (dollars)	2	4	6	8	10

Writing the values in order gives the list

$$2, 4, 6, 8, 10.$$

An **entry** is one value in a list. A **pattern** is a rule connecting the entries. Here each entry increases by \$2.

If Mina continues for ten days, the list is

$$2, 4, 6, 8, 10, 12, 14, 16, 18, 20.$$

Once the pattern is clear, we can shorten it:

$$2, 4, 6, \ldots, 20.$$

The symbol $\ldots$ is an **ellipsis**, or dots. It means that predictable middle entries were left out.

For example,

$$18, 15, 12, 9, 6, 3 \quad \longrightarrow \quad 18, 15, 12, \ldots, 3.$$

The dots do not create the pattern: the rule must already be clear.

For example,

$$2, 4, \ldots, 16$$

is ambiguous. It could mean

$$2, 4, 6, 8, 10, 12, 14, 16 \qquad \text{or} \qquad 2, 4, 8, 16.$$

If more than one reasonable pattern is possible, state the intended rule or show more entries.

Say it in words. Read $5, 10, 15, \ldots, 30$ as "5, 10, 15, and the same add-5 pattern continues up to 30."

Checkpoint. A list records values in order. A pattern explains how the values are connected, and $\ldots$ safely removes predictable middle entries.

1.2 Giving Quantities Short Names (C0)

A small shop sells cups of fruit for \$3 each. Repeating "number of cups sold" in every sentence is clumsy. A **variable** is a symbol used as a short name for a quantity:

$$\text{number of cups sold} \quad \longrightarrow \quad Q = \text{number of cups sold}.$$

The meaning of Q stays fixed, but its value can change. For example,

$$Q = 2 \text{ cups} \qquad \text{or} \qquad Q = 5 \text{ cups}.$$

Now define three quantities:

$$P = \text{price of one cup}, \qquad Q = \text{cups sold}, \qquad R = \text{money received}.$$

Then

$$R = P \times Q.$$

For $P = 3$ dollars per cup and $Q = 4$ cups,

$$R = 3 \times 4 = 12 \text{ dollars}.$$

R is **revenue**: money received from sales. It is not necessarily profit.

Say it in words. Read $R = P \times Q$ as "money received equals price per cup times the number of cups sold."

A letter has no automatic meaning. Its definition tells us what it represents.

Checkpoint. A variable's meaning stays fixed after it is defined, although its value may change. Next, subscripts will let one letter name several related quantities.

2 From Labelled Quantities to General Collections

Problems:

2.1 Organizing Related Quantities with Subscripts (C0)

The shop sells apples, bread, and milk. Their prices form a **collection** of related quantities:

$$\text{price of apples}, \quad \text{price of bread}, \quad \text{price of milk}.$$

Using only P would give the same name to all three prices. We need a label that tells us which price we mean.

First, we could use word labels:

$$P_{\text{apples}}, \quad P_{\text{bread}}, \quad P_{\text{milk}}.$$

The small lower label is called a **subscript**.

We can make the labels shorter:

$$P_a, \quad P_b, \quad P_m,$$

where

$$a = \text{apples}, \qquad b = \text{bread}, \qquad m = \text{milk}.$$

Thus P_b means "the price of bread."

This works well for a few products. But suppose the shop sells many products:

$$\text{apples, bread, milk, rice, cheese, juice, eggs}, \ldots$$

Choosing and remembering a different letter for every product becomes inconvenient.

Instead, number the products:

Product	apples	bread	milk
Number	1	2	3

Then

$$P_a, P_b, P_m \quad \longrightarrow \quad P_1, P_2, P_3.$$

Now the same idea works for collections of any size:

$$P_1, P_2, P_3, P_4, \ldots$$

Each subscript tells us which member of the collection we mean. For example,

$$P_2 = \text{price of product 2}.$$

If product 2 is bread and bread costs \$3, then

$$P_2 = 3 \text{ dollars}.$$

Why use numbers? Letters such as a, b, m can be useful when there are only a few clearly named objects. Numbers are easier to continue when the collection becomes large:

$$P_1, P_2, P_3, \ldots, P_{20}.$$

We do not need to invent 20 different letter labels.

A subscript is a label, not an arithmetic operation:

$$\begin{aligned} P_2 &= \text{the member of the collection labelled 2,} \\ P^2 &= P \times P, \\ 2P &= 2 \times P. \end{aligned}$$

Say it in words. Read P_3 as "P sub three." The 3 identifies which P we mean; it does not mean multiply by 3.

2.2 Describing Any Collection Size with n (C0)

Suppose a shop sells three products. We can name their prices

$$P_1, P_2, P_3.$$

If the shop sells five products, we need

$$P_1, P_2, P_3, P_4, P_5.$$

If it sells eight products, we need

$$P_1, P_2, P_3, P_4, P_5, P_6, P_7, P_8.$$

Each list works, but only for that particular shop.

In economics, the number of items often changes. One shop may sell 5 products, another may sell 20, and a larger store may sell hundreds. We want one notation that can describe the prices without choosing the number of products first.

For a shop with 8 products, we can shorten the list:

$$P_1, P_2, P_3, \ldots, P_8.$$

Here P_8 tells us exactly where the collection ends.

Now imagine that we only say:

"The shop sells some number of products."

We do not yet know whether that number is 5, 8, 20, or something else. We need a symbol for that unknown or changing number.

Define

$$n = \text{number of products.}$$

Then the last price is

$$P_n.$$

So the complete collection can be written

$$P_1, P_2, P_3, \ldots, P_n.$$

This notation means:

$$\underbrace{P_1}_{\text{price 1}}, \underbrace{P_2}_{\text{price 2}}, \underbrace{P_3}_{\text{price 3}}, \ldots, \underbrace{P_n}_{\text{last price}}.$$

The letter n does not mean one particular number. Its value depends on the situation.

For example,

Number of products	n	Price collection
3	3	P_1, P_2, P_3
5	5	$P_1, P_2, \ldots, P_5$
8	8	$P_1, P_2, \ldots, P_8$

All three cases can therefore be described at once by

$$P_1, P_2, \ldots, P_n.$$

This is called a **generalization**. Instead of writing a new rule for every possible number of products, we write one rule that works for any collection size.

The same idea appears in many financial and economic collections. For example, if a family has some number of monthly expenses, we could write

$$E_1, E_2, \ldots, E_n,$$

where n would tell us how many expenses are in the collection.

If a company has some number of products with revenues

$$R_1, R_2, \ldots, R_n,$$

then R_n means the revenue of the last product in the numbered collection.

Say it in words. Read

$$P_1, P_2, \ldots, P_n$$

as "the prices from product 1 up to product n, where n is the total number of products."

Checkpoint. A fixed ending such as P_8 describes one particular collection. Replacing the final number by n lets us describe a collection whose size may change:

$$P_1, P_2, \ldots, P_n.$$

Here n tells us how many members the collection has. Next, we need notation for referring to an arbitrary member somewhere inside that collection.

2.3 Selecting Any One Entry with i (C0)

Suppose a shop sells four products. Their prices are

$$P_1, P_2, P_3, P_4.$$

Each subscript tells us which product we mean:

$$P_1 = \text{price of product 1}, \qquad P_2 = \text{price of product 2},$$

and so on.

If we already know the product number, this notation is enough. For example,

$$P_3$$

means the price of product 3.

But sometimes we want to talk about *one product without choosing its number yet.*

For example, imagine that we want to say:

"Take the price of whichever product we are considering."

Writing P_1 would choose product 1. Writing P_2 would choose product 2. We need a symbol that can stand for the chosen position.

Define

$$i = \text{the position of the product we are considering.}$$

Then write

$$P_i.$$

This means

$$P_i = \text{price of product } i.$$

The value of i tells us which price is selected:

i	P_i
1	P_1
2	P_2
3	P_3
4	P_4

So if

$$i = 2,$$

then

$$P_i = P_2.$$

If

$$i = 4,$$

then

$$P_i = P_4.$$

The symbol P_i therefore does not name one fixed price. It means "the price at the position selected by i."

Why is this useful? Suppose the same four products also have quantities sold:

$$Q_1, Q_2, Q_3, Q_4.$$

The revenue from product 1 is

$$P_1 Q_1.$$

The revenue from product 2 is

$$P_2 Q_2.$$

The revenue from product 3 is

$$P_3 Q_3.$$

Instead of writing a separate expression for every product, we can write

$$P_i Q_i.$$

This means

$$\text{price of product } i \times \text{quantity sold of product } i.$$

For example, if $i = 3$, then

$$P_i Q_i = P_3 Q_3.$$

The index i lets the same notation work for whichever product we choose.

Say it in words. Read

$$P_i$$

as "P sub i: the price of product i."

Read

$$P_i Q_i$$

as "the price of product i times the quantity sold of product i."

Checkpoint. A fixed subscript such as

$$P_3$$

selects one known position.

A variable subscript such as

$$P_i$$

lets the position change.

Thus i answers the question:

"Which entry are we talking about?"

2.4 Describing a Whole Indexed Collection (C0)

The notation

$$P_i$$

lets us talk about one selected price.

Now suppose we want to describe *all* the prices in a collection.

For a shop with four products, the complete list is

$$P_1, P_2, P_3, P_4.$$

Since P_i means "the price at position i," we could describe the same collection by saying:

"Take P_i for $i = 1, 2, 3, 4$."

That instruction produces

$$P_1, P_2, P_3, P_4.$$

A compact notation for this is

$$\{P_i\}_{i=1}^{4}.$$

It means:

"the collection of P_i values obtained as i goes from 1 to 4."

So

$$\{P_i\}_{i=1}^{4} = \{P_1, P_2, P_3, P_4\}.$$

The notation has three important parts:

$$\underbrace{P_i}_{\text{what kind of entry}} \qquad \underbrace{i = 1}_{\text{where we start}} \qquad \underbrace{4}_{\text{where we stop}} \quad .$$

For example,

$$\{R_i\}_{i=1}^{3}$$

means

$$\{R_1, R_2, R_3\},$$

the revenues of products 1 through 3.

Similarly,

$$\{C_i\}_{i=1}^{5}$$

means

$$\{C_1, C_2, C_3, C_4, C_5\},$$

the costs of products 1 through 5.

Why do we need n? The notation above still requires us to know the final number.

For four products,

$$\{P_i\}_{i=1}^{4}.$$

For seven products,

$$\{P_i\}_{i=1}^{7}.$$

For twelve products,

$$\{P_i\}_{i=1}^{12}.$$

But in economics, we often want one description that works even when the number of products changes.

Suppose one shop has 4 products, another has 7, and another has 20. Instead of writing a different ending every time, define

$$n = \text{number of products.}$$

Then write

$$\{P_i\}_{i=1}^{n}.$$

This means

$$\{P_1, P_2, \ldots, P_n\}.$$

Narratively,

$$\{P_i\}_{i=1}^{n}$$

means:

> "the collection of all product prices, starting with product 1 and continuing through product n."

Another way to say it is:

> "take P_i once for every product position from 1 through n."

Here the two letters have different jobs:

$$i = \text{which position we are currently using,}$$

while

$$n = \text{how many positions exist.}$$

If there are 5 products, then

$$n = 5,$$

and

$$\{P_i\}_{i=1}^{n} = \{P_i\}_{i=1}^{5} = \{P_1, P_2, P_3, P_4, P_5\}.$$

As the collection is generated, i takes the values

$$1, 2, 3, 4, 5.$$

In general,

$$i = 1, 2, \dots, n,$$

or more compactly,

$$1 \leq i \leq n.$$

One entry versus the whole collection. It is important to distinguish

$$P_i$$

from

$$\{P_i\}_{i=1}^{n}.$$

The first means

$$\boxed{P_i = \text{one selected price}},$$

while the second means

$$\boxed{\{P_i\}_{i=1}^{n} = \text{the whole collection of prices}}.$$

For example, if $n = 4$,

$$P_i$$

could be P_1, P_2, P_3, or P_4 depending on i.

But

$$\{P_i\}_{i=1}^{4}$$

means all four:

$$\{P_1, P_2, P_3, P_4\}.$$

Say it in words. Read

$$\{P_i\}_{i=1}^{6}$$

as "the collection of prices P_i for positions 1 through 6."

Read

$$\{P_i\}_{i=1}^{n}$$

as "the collection of prices P_i for positions 1 through n, where n is the number of products."

Checkpoint. The notation

$$P_i$$

describes one entry whose position may change.

The notation

$$\{P_i\}_{i=1}^{n}$$

describes the whole collection by letting i move through

$$1, 2, \dots, n.$$

Thus,

$$i = \text{which entry}, \qquad n = \text{how many entries}.$$

Next, some economic situations require two changing positions at the same time. This motivates using two indices, such as i and j.

2.5 Using Two Labels for a Table (C0)

The shop records how many items are sold for three products on four days. Define

$$i = \text{product row}, \qquad j = \text{day column}, \qquad S_{ij} = \text{items of product } i \text{ sold on day } j.$$

Here S means quantity sold, in items. The first subscript chooses a product. The second chooses a day.

	$j=1$ Mon.	$j=2$ Tue.	$j=3$ Wed.	$j=4$ Thu.
$i=1$ apples	$S_{11}=3$	$S_{12}=5$	$S_{13}=4$	$S_{14}=6$
$i=2$ bread	$S_{21}=2$	$S_{22}=4$	$S_{23}=3$	$S_{24}=5$
$i=3$ milk	$S_{31}=4$	$S_{32}=3$	$S_{33}=5$	$S_{34}=2$

To find S_{23}, trace row $i = 2$ (bread), then column $j = 3$ (Wednesday). The cell contains 3, so $S_{23} = 3$ items: 3 bread items were sold on Wednesday.

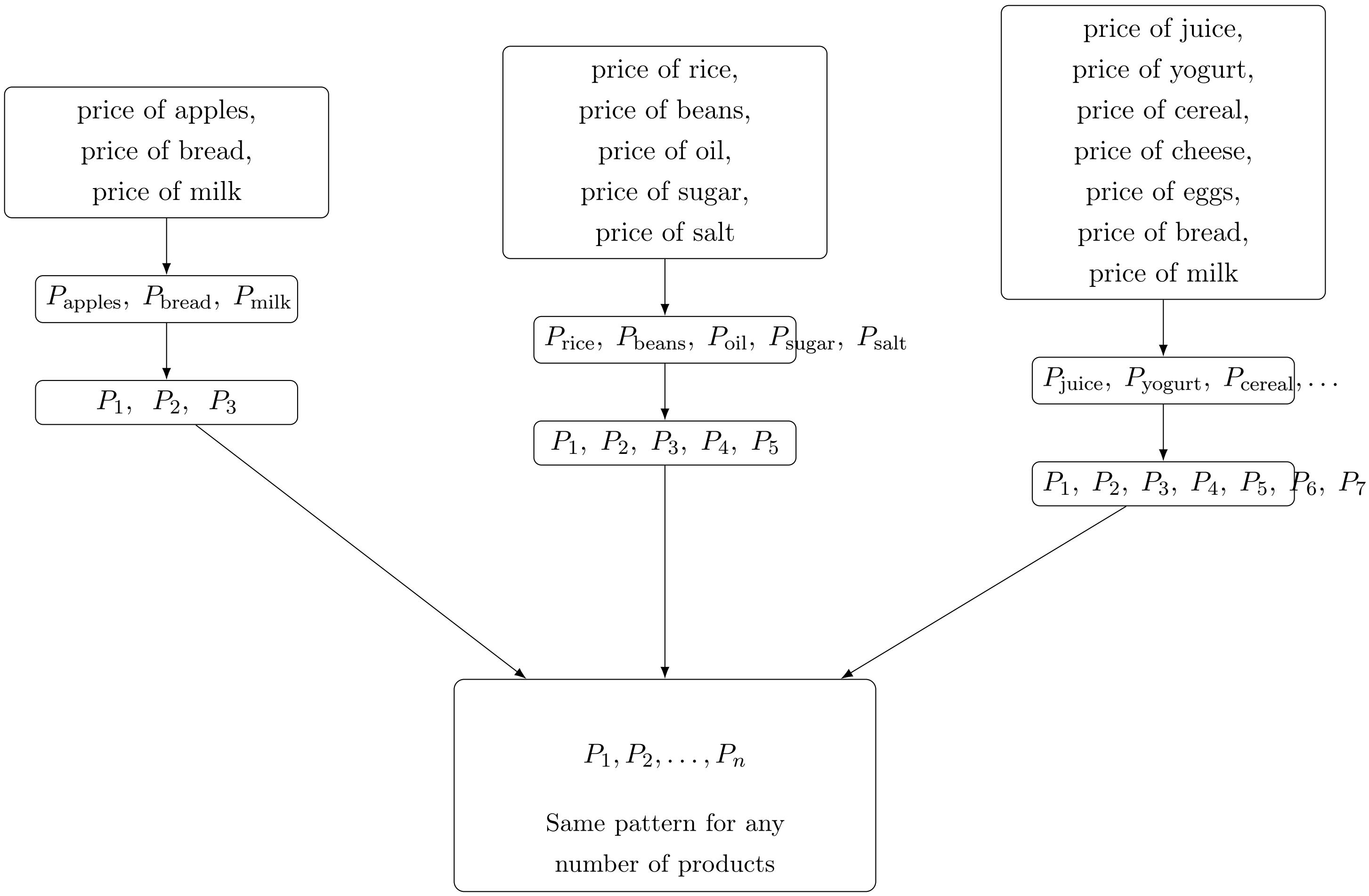
Check with substitution: if $i = 3$ and $j = 2$, then $S_{ij} = S_{32} = 3$ items. The labels can change independently. Keeping $i = 3$ and changing j follows milk across days; keeping $j = 2$ and changing i follows products down Tuesday's column.

Say it in words. Read S_{ij} as "the number of items of product i sold on day j." The letters have these jobs only because we defined them that way.

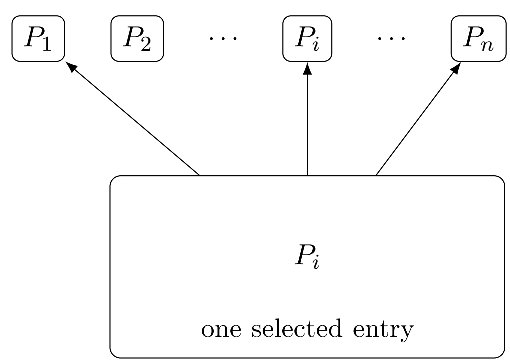
Checkpoint. In S_{ij}, i and j choose different directions and may change separately. We are ready to join the whole notation chain.

2.6 Putting the Notation Together (C0)

Follow each arrow slowly. No step changes what a price means.



Think of the general list as many possible entries:



i chooses which position we mean

Say it in words. $P_1, P_2, \dots, P_n$ means the whole price collection. P_i means one selected price. S_{ij} means one sales count selected by both a product and a day.

Checkpoint. Compact notation is useful only when every letter, label, unit, and range can be explained in words.

3 Review and Transfer

Problems:

3.1 Notation Reference (C0)

Notation	How to read it	Its job	Economic example
Q	"Q"	Names one defined quantity	cups sold
$R = P \times Q$	"R equals P times Q"	Relates defined quantities	revenue equals price times quantity
$\dots$	"and the clear pattern continues"	Omits predictable entries	$2, 4, 6, \dots, 20$ dollars saved
P_2	"P sub two"	Selects labelled member 2	price of product 2
$P_1, \dots, P_n$	"prices 1 through n"	Names a collection of size n	all n product prices
P_i	"P sub i"	Selects any allowed price	price of product i
$1 \le i \le n$	"i is from 1 through n"	Gives allowed whole-number positions	any product position
S_{ij}	"S sub i j"	Selects a row and a column	items of product i sold on day j

Checkpoint. The table is a reminder, not a list of automatic meanings. In a new story, define every symbol again.

3.2 Describing Any Collection Size with n (C0)

Suppose three small shops sell different numbers of products.

Shop A

$$P_1, P_2, P_3$$

The collection ends at P_3.

Shop B

$$P_1, P_2, P_3, P_4, P_5$$

The collection ends at P_5.

Shop C

$$P_1, P_2, P_3, P_4, P_5, P_6, P_7$$

The collection ends at P_7.

The same notation pattern appears each time:

$$P_1, P_2, P_3, \dots$$

but the place where the list stops changes.

The problem. Suppose we want to describe the prices of *any shop*, without deciding first whether it sells 3, 5, 7, or 20 products.

We need a symbol for

"the number of products in this shop."

Define

$$n = \text{number of products.}$$

Then the last product is product n, so its price is

$$P_n.$$

Therefore the whole price collection can be written

$$P_1, P_2, \ldots, P_n.$$

This one expression can represent collections of different sizes.

If $n = 3$,

$$P_1, P_2, P_3.$$

If $n = 5$,

$$P_1, P_2, P_3, P_4, P_5.$$

If $n = 7$,

$$P_1, P_2, P_3, P_4, P_5, P_6, P_7.$$

When n changes, the number of entries changes.

What does n do? The symbol n has one important job:

$$\boxed{n = \text{how many entries are in the collection.}}$$

Because the products are numbered starting at 1, n is also the number on the final product.

For example, if

$$n = 6,$$

then

$$P_1, P_2, \ldots, P_n$$

becomes

$$P_1, P_2, P_3, P_4, P_5, P_6,$$

so

$$P_n = P_6.$$

Here P_n does *not* mean any product price. It means the price of the final product.

Why is this a generalization? **Fixed size**

$$P_1, P_2, \ldots, P_5$$

This always describes exactly 5 prices.

General size

$$P_1, P_2, \ldots, P_n$$

This can describe 3, 5, 20, 100, or another number of prices.

Using a symbol instead of one fixed number lets the same notation describe many possible situations. This is a **generalization**.

The same idea works for other economic quantities:

$$C_1, C_2, \ldots, C_n$$

can describe the costs of n products.

$$R_1, R_2, \ldots, R_n$$

can describe the revenues of n products.

Say it in words. Read

$$P_1, P_2, \ldots, P_n$$

as

"the prices of products 1, 2, and so on, up to product n."

Here n tells us how many products there are.

Checkpoint. A fixed ending such as

$$P_8$$

describes one particular collection size, while

$$P_n$$

allows the size to change.

Thus,

$$\boxed{n = \text{how many entries are in the collection.}}$$

Next, we need a way to talk about one entry without choosing its position in advance.

3.3 Selecting Any One Entry with i (C0)

Suppose a shop sells five products with prices

$$P_1, P_2, P_3, P_4, P_5.$$

If we want the price of product 2, we write

$$P_2.$$

If we want the price of product 4, we write

$$P_4.$$

The number in the subscript tells us which entry we selected.

The problem. Now suppose we want to write a statement about one product, but we do not want to choose its position yet.

We might mean

$$P_1, \quad P_2, \quad P_3, \quad P_4, \quad \text{or} \quad P_5.$$

Instead of choosing a fixed number, define

$$i = \text{selected product position.}$$

Then

$$P_i$$

means

the price of product i.

The value of i tells us which particular price is selected.

If $i = 1$,

$$P_i = P_1.$$

If $i = 3$,

$$P_i = P_3.$$

If $i = 5$,

$$P_i = P_5.$$

So P_i does not mean one fixed price. It means

"the price at whichever position i currently selects."

Why is this useful? Suppose the same five products also have quantities sold:

$$Q_1, Q_2, Q_3, Q_4, Q_5.$$

The revenue rule repeats for every product:

Product 1:

$$P_1 Q_1.$$

Product 2:

$$P_2 Q_2.$$

Product 3:

$$P_3 Q_3.$$

Instead of writing the same relationship separately for every product, we can write

$$P_i Q_i.$$

This means

$$\text{price of product } i \times \text{quantity sold of product } i.$$

For example:

If $i = 2$,

$$P_i Q_i = P_2 Q_2.$$

If $i = 5$,

$$P_i Q_i = P_5 Q_5.$$

The symbol i lets one expression work for whichever product we choose.

n and i have different jobs. Now consider a shop with

$$P_1, P_2, \ldots, P_n.$$

We have

$$n = \text{number of products}$$

and

$$i = \text{selected product position}.$$

$$\boxed{n = \text{how many entries exist}}$$

n determines where the collection ends.

$$\boxed{i = \text{which entry we select}}$$

i determines which member we are talking about.

For example, suppose

$$n = 6.$$

Then

$$P_1, P_2, P_3, P_4, P_5, P_6$$

is the complete collection, while i may take any of the values

$$1, 2, 3, 4, 5, 6.$$

Therefore,

$$1 \leq i \leq 6.$$

More generally,

$$1 \leq i \leq n.$$

This means:

"i may be any whole-number position from 1 through n."

A particularly important contrast is:

$$P_n$$

is the price of the **final** product.

$$P_i$$

is the price of **whichever product** i **selects**.

For example, if

$$n = 6, \qquad i = 3,$$

then

$$P_n = P_6, \qquad P_i = P_3.$$

Say it in words. Read

$$P_i$$

as "P sub i: the price of product i."

Read

$$1 \leq i \leq n$$

as "i can be any whole-number position from 1 through n."

Checkpoint.

$\boxed{n: \text{ How many entries are there?}}$

$\boxed{i: \text{ Which entry are we talking about?}}$

Thus,

$$P_1, P_2, \ldots, P_n$$

describes the available positions, while

$$P_i$$

lets us refer to one of them without choosing it in advance.

Next, we can let i move through every allowed position to describe the entire indexed collection.